Technical manual: FaceAttend EDU

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Technical manual: FaceAttend EDU

1. Introduction

FaceAttend EDU is an educational attendance management platform that replaces manual attendance taking with an automated process based on *facial and fingerprint biometric identification*. The system validates the identity of students and teachers, manages the complete academic cycle (programs, courses, schedules, sessions and enrollments) and issues auditable, traceable attendance records.

The solution is conceived as a formal software engineering project: every design decision is documented by means of *Architecture Decision Records (ADRs)*, the services are organized by *bounded contexts* under DDD, and quality is verified against recognized standards (ISO/IEC 25010 for quality attributes, IEEE 829 and ISTQB for test documentation) (Institute of Electrical and Electronics Engineers [IEEE], 1998; International Organization for Standardization & International Electrotechnical Commission [ISO & IEC], 2011; International Software Testing Qualifications Board, 2018).

2. Objective

To develop and implement a technology platform that makes it possible to:

1. *Record and verify educational attendance biometrically* (face and fingerprint), eliminating identity impersonation ("proxy attendance") and manual recording errors.
2. *Manage the complete academic cycle*: programs, courses, schedules, scheduled sessions and student enrollment.
3. *Guarantee security and traceability*: opaque-session authentication (UUID sessionId validated against ms-identity), RBAC authorization, auditing on all entities and layered network isolation.
4. *Demonstrate software quality under formal standards*, with objective metrics reported by a dedicated microservice (ms-quality) and test documentation compliant with IEEE 829 / ISTQB.

5. *Operate reproducibly* by means of infrastructure as code (Docker Compose) and database as code (Liquibase + versioned SQL scripts) (Docker Inc., n.d.).

3. Technical scope

In scope:

- *Polyglot modular monolith* architecture with DDD (8 domain bounded contexts + API Gateway) and *hexagonal architecture* (ports and adapters) inside each service.
- *9 microservices* specialized in Java 21, TypeScript (Node 24) and Python 3.12, orchestrated with Docker Compose.
- *Kong 3.6 API Gateway* (DB-less mode) as the only public entry point, with rate limiting via Redis (Redis Ltd., n.d.).
- Persistence: *PostgreSQL 17* multi-schema (database-per-service strategy), *MongoDB 7* for biometric embeddings, *Redis 7* for caching/rate limiting and *Kafka 3.8 (KRaft)* as the asynchronous event bus (MongoDB Inc., n.d.).
- Versioned migrations with *Liquibase 4.29*; documentary schema modeling in *DBML*.
- *Web* and *Mobile* clients built with React Native + Expo (the web app exported with `expo export --platform web` and served by Nginx), organized under the *MVVM* pattern (Expo, n.d.).
- Network segmentation into *3 zones* (`edge`, `app`, `data`) as an infrastructure security control.
- Quality module with ISO/IEC 25010, ISTQB, IEEE 829 and ISO 29110 reports, and a catalog of error/quality codes (International Organization for Standardization & International Electrotechnical Commission [ISO & IEC], 2016).

Out of scope (for now):

- Integration with physical on-site fingerprint capture hardware (work is done against embedding APIs).
- Managed cloud deployment (Kubernetes/Terraform); the current target environment is local/containers.

- Billing, payments or administrative modules unrelated to attendance and the academic cycle.

4. Technical description of the system (solution summary)

FaceAttend EDU implements a *layered client-server* pattern with synchronous (REST) and asynchronous (event-based) communication:

- The *clients* (Web and Mobile, both React Native/Expo with MVVM logic) never access the services or the databases directly: every request passes through *Kong*, which resolves routes, applies CORS and rate limiting, and requires the `sessionId` of an active session (validated against `ms-identity`). Each microservice validates the session again and evaluates the RBAC rules of `ms-authorization`.
- At the application layer, each microservice encapsulates a bounded context: `identity` (users and authentication), `authorization` (roles/permissions), `academic` (programs, courses, enrollments), `scheduling` (schedules and sessions), `attendance` (attendance recording), `biometric` (facial/fingerprint templates and vector matching in MongoDB), `configuration` (system parameters) and `notification` (alerts and email delivery). `quality` exposes system verification metrics.
- The *data layer* remains isolated in its own Docker network: PostgreSQL per context (no foreign keys between contexts — only logical UUIDs), MongoDB for embeddings, Kafka for domain events and Redis as gateway support.
- The critical attendance flow combines both worlds: biometric verification → validation of the current session and of the enrollment → transactional write of the record → publication of the event on the `attendance-events` topic, best-effort and outside the transaction.
- Startup is deterministic and verifiable: PostgreSQL → Liquibase migrations → microservices → ms-quality → Kong + Redis → frontend, chained together with Docker Compose *healthchecks*.

5. Technologies used

5.1 Languages and frameworks (by component)

Table 1

5.1 Languages and frameworks (by component)

Component	Language	Exact version	Main framework / library	Version
01-ms-identity	Java	21 (Eclipse Temurin)	Spring Boot	4.1.2
02-ms-authorization	Java	21 (Eclipse Temurin)	Spring Boot	4.1.2
04-ms-scheduling	Java	21 (Eclipse Temurin)	Spring Boot	4.1.2
05-ms-attendance	Java	21 (Eclipse Temurin)	Spring Boot	4.1.2
03-ms-academic	TypeScript	^5.4.0 (Node 24)	Fastify + pg	^5.12.7 / ^8.11.0
07-ms-configuration	TypeScript	^5.4.0 (Node 24)	Fastify + pg	^5.12.7 / ^8.11.0
09-ms-quality	TypeScript	^5.4.0 (Node 24)	Fastify + pg	^5.12.7 / ^8.11.0
06-ms-biometric	Python	^3.12	FastAPI + Uvicorn + OpenCV	^0.162.0 / ^0.54.0 / ^4.9.0
08-ms-notification	TypeScript	^5.4.0 (Node 24)	Fastify + pg + nodemailer	^5.12.7 / ^8.11.0 / ^10.2.1
Web frontend	JavaScript/TypeScript	TS ~6.0.3 · React 19.2.3	React Native + Expo (`react-native-web`)	RN 0.86.3 / Expo ^57.0.20
Mobile frontend	JavaScript/TypeScript	TS ~5.9.2 · React 19.2.3	React Native + Expo	RN 0.86.3 / Expo ~57.0.0

Source. Own elaboration.

5.2 Databases and messaging

Table 2

5.2 Databases and messaging

Technology	Role	Exact version
PostgreSQL	Multi-schema relational DB (database-per-context)	17-alpine
MongoDB	Store for biometric embeddings (vector search)	7

Technology	Role	Exact version
Redis	Gateway caching and rate limiting	7-alpine
Apache Kafka (KRaft, without Zookeeper)	Asynchronous event bus between contexts	3.8.0
Liquibase	Schema migrations (database-as-code)	4.29.0
DBML	Documentary modeling of the schema (`faceattend_edu_mr_v4.dbml`)	—

Source. Own elaboration.

5.3 Infrastructure, gateway and deployment

Table 3

5.3 Infrastructure, gateway and deployment

Technology	Role	Exact version
Kong (OSS, DB-less)	API Gateway: routing, CORS, rate limiting	3.6
Docker Compose	Local orchestration with 3 segmented networks	own declarative stack
Nginx	Static server for the web bundle (`expo export`)	1.27-alpine
Node.js	Runtime of the TypeScript services (`node:24-slim`) and frontend build (`node:24-alpine`)	24
Maven (mvnw wrapper)	Build of the Java services	3.9 (`maven:3.9-eclipse-temurin-21-alpine` image)
Poetry	Python dependency management (biometric)	python ^3.12

Source. Own elaboration.

5.4 Security

Table 4

5.4 Security

Technology	Role	Version
Spring Security	Authentication/authorization in the Java services	included in Spring Boot 4.1.2
Opaque session	UUID `sessionId` persisted in `identity.user_session`; no JWT library	in-house (ms-identity)

Source. Own elaboration.

5.5 Version control

Table 5

5.5 Version control

Tool	Version	Use in the project
Git	2.39.5	Monorepo with branches per integration cycle (`develop` as the working branch); conventional commits (`chore(compose): …`, `feat(ms-x): …`)
GitHub	SaaS platform	Remote hosting, backup and review of the repository

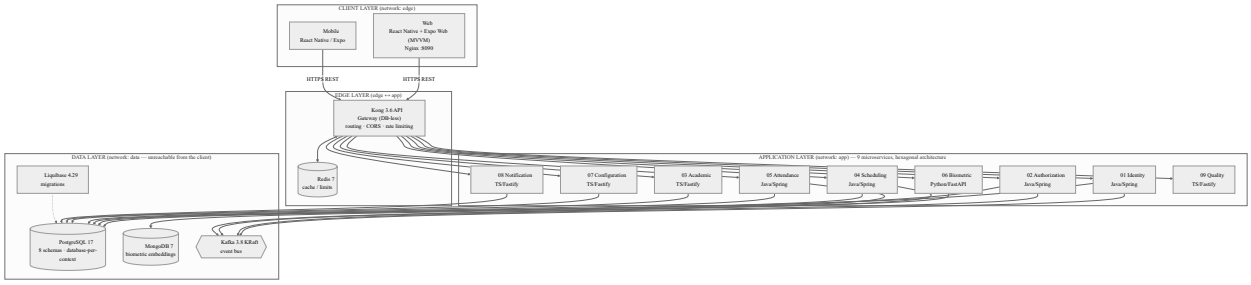
Source. Own elaboration.

6. System architecture

6.1 Layer diagram (client-server)

Figure 1

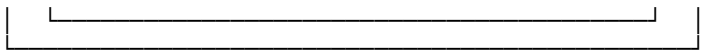
6.1 Layer diagram (client-server)



Source. Own elaboration.

6.2 Internal structure of each service (hexagonal)

```
┌──────────────────────────────────────────────────┐
│  domain/          Entities, value objects,       │  ← pure business rules,
│                   ports (interfaces)             │    no external dependencies
│  ┌──────────────────────────────────────────┐    │
│  │ application/  Use cases that orchestrate │    │  ← use cases
│  │               the domain                 │    │
│  │  ┌──────────────────────────────────┐    │    │
│  │  │ infrastructure/adapters/         │    │    │  ← HTTP controllers,
│  │  │  http · persistence · messaging  │    │    │    JPA/pg/Mongo,
│  │  └──────────────────────────────────┘    │    │    Kafka producers/consumers
│  └──────────────────────────────────────────┘    │
└──────────────────────────────────────────────────┘
```



6.3 Explanation of the flow

Synchronous flow (REST through Kong):

1. *Authentication:* the client sends credentials to `POST /api/v1/auth/login` → Kong routes to *ms-identity* → validates the credentials (BCrypt) against PostgreSQL → creates a row in `identity.user_session` and returns its `sessionId` (opaque UUID) → *ms-authorization* resolves roles and RBAC permissions on every request.
2. *Biometric enrollment:* face/fingerprint capture → *ms-biometric* generates the embedding (vector) and stores it in *MongoDB*; the template is linked to the student via UUID.
3. *Attendance recording:* the teacher opens a session (*ms-scheduling*) → the student is verified → *ms-attendance* coordinates: biometric *matching* (ms-biometric), temporal validity of the session (ms-scheduling) and active enrollment (ms-academic) → writes the record transactionally to PostgreSQL.

Asynchronous flow (Kafka events):

- Every relevant action publishes a domain event on its entity topic (`attendance-events`, `identity-events`, `role-events`, and so on; see Section 11.8).
- Publication does not block the transactional flow: if the broker does not respond, the failure is logged as a warning and the operation continues.
- *Current consumer:* `ms-notification` (TypeScript, `kafkajs`) consumes `attendance.justification.events` (group `notification-service`) and `scheduling.class-session.closed` (group `notification-service-class-session`) to build alerts, which it sends by SMTP. The other topics have no consumers yet; they can be added without touching the producers.

Cross-cutting rules:

- Only *logical UUIDs* travel between bounded contexts (no real FKs between schemas) → low coupling.
- Standard auditing on all tables: soft-delete, timestamps and optimistic locking.
- Security by network: the client can only reach Kong; Kong can only reach the `app` layer; only the `app` layer can reach the `data` network.

Orchestrated startup order (healthchecks): PostgreSQL → Liquibase migrations per context → microservices → ms-quality → Kong + Redis → Web frontend.

7. Project structure

7.1 Folder tree

```
/workspace
├── docker-compose.yml                  # General orchestration: edge/app/data
networks, healthchecks
├── COMPOSE.md                          # Documentation of the Docker Compose
environment
│
├── front-end/
│   ├── Web/                            # React Native + Expo app exported to web
(react-native-web)
│   │   ├── Dockerfile                  # Build node:20 → expo export → nginx:1.27
│   │   ├── nginx.conf                  # Static serving + SPA fallback
│   │   └── src/
│   │       ├── api/                    # Base HTTP client towards Kong
│   │       ├── config/                 # Environment configuration (URLs, variables)
│   │       ├── context/                # Global contexts (Auth, Theme, Responsive)
│   │       ├── core/                   # Cross-cutting: constants, hooks, storage,
theme, utils
│   │       ├── models/                 # Types and data structures per domain
│   │       ├── navegation/             # React Navigation: public/authenticated
stacks + deep linking
│   │       ├── services/api/           # One module per microservice (authApi,
academicApi, ...)
│   │       ├── view/                   # VIEW (V): components/ and screens/
(authorized/)
│   │       └── viewmodels/             # VIEWMODEL (VM): hooks with presentation
logic
│   └── Mobile/                         # React Native + Expo mobile app
│       ├── src/
│       │   ├── api/ · services/        # HTTP access to Kong
│       │   ├── assets/images/          # Static resources
│       │   ├── hooks/ · utils/         # Hooks and utilities (locales/i18n)
│       │   ├── models/                 # Types per context: identity, academic,
attendance,
│       │   │                           #   scheduling, authorization, notification,
...
│       │   ├── navigations/            # React Navigation
│       │   └── storage/                # Local persistence of the sessionId
(SecureStore)
│       ├── scripts/                    # Development utilities
```

```
│       └── seed/                          # Seed data for mobile testing
│
├── back-end/                              # Microservices + gateway (DDD + hexagonal)
│   ├── 01-ms-identity/                    # Java 21 / Spring Boot — users,
authentication and sessions
│   │   └── src/main/java/com/faceattend_edu/identity_service/
│   │       ├── domain/                    # Entities and ports
│   │       ├── application/               # Use cases
│   │       ├── adapter/                   # REST controllers, JPA, Kafka
│   │       ├── config/ · shared/          # Cross-cutting (security, exceptions, audit)
│   ├── 02-ms-authorization/               # Java — RBAC roles and permissions
│   ├── 03-ms-academic/                    # TS/Fastify — programs, courses, enrollments
│   │   └── src/{domain/{entities,ports}, application/use-cases,
│   │            infrastructure/{http,persistence,db,messaging}}
│   ├── 04-ms-scheduling/                  # Java — schedules and sessions
│   ├── 05-ms-attendance/                  # Java — attendance recording
│   ├── 06-ms-biometric/                   # Python/FastAPI — face/fingerprint,
embeddings → MongoDB
│   │   └── {domain/{entities,value_objects,ports}, infrastructure/{web,persistence},
│   │        tests/{unit,support}}
│   ├── 07-ms-configuration/               # TS/Fastify — system parameters
│   ├── 08-ms-notification/                # TypeScript/Fastify — alerts + SMTP email
│   │   └── {cmd/, internal/{domain/port, application/usecase,
│   │            infrastructure/{http,postgres,config}}, migrations/}
│   ├── 09-ms-quality/                     # TS/Fastify — ISO/IEC 25010 metrics and
reports
│   ├── 99-api-gateway/kong/kong.yml       # DB-less definition: routes, upstreams,
plugins
│   ├── quality/quality-codes.md           # Catalog of quality/error codes
│   ├── docker-compose.yml                 # Backend stack (Kong, Kafka, Mongo, Redis,
MSs)
│   └── *.md                               # Formal documents: QUALITY.md, ISO25010,
│                                          #   IEEE829 (plan/cases/incidents/summary),
│                                          #   ISTQB, ISO29110, SERVICES.md, DATABASE.md
│
├── database/                              # Database-as-code, one directory per context
│   ├── 01-ms-identity-db/ … 08-ms-notification-db/
│   │   ├── 01-ddl/                        # Tables, indexes, triggers
│   │   ├── 02-dml/                        # Seeds (initial data)
│   │   ├── 03-dcl/                        # Roles, grants, security
│   │   ├── 04-tcl/                        # Transaction control
│   │   ├── 05-rollbacks/                  # Reversible scripts
│   │   └── changelog/                     # Liquibase changelogs (changelog.xml)
│   ├── database-init/                     # PostgreSQL bootstrap (schemas and users)
│   ├── scripts/init-multidb.sql           # Multi-database initialization
│   ├── docker-compose.yml                 # postgres:17-alpine + liquibase:4.29.0
│   ├── faceattend_edu_mr_v4.dbml          # Documentary relational model (DBML)
│   └── CONVENCIONES.md · ESTRUCTURA.md · MODELO.md · SEEDS.md
│
└── tools/
    └── seed-api-test-data/                # Test data loader consuming the real API
```

7.2 Explanatory table of the main folders

Table 6

7.2 Explanatory table of the main folders

Folder	What it does
`docker-compose.yml` (root)	Single entry point of the stack: defines the 3 isolated networks (`edge`, `app`, `data`), volumes and the startup chain with healthchecks.
`front-end/Web/src/view`	View (V) layer of MVVM: reusable components and screens (public and `authorized/`).
`front-end/Web/src/viewmodels`	ViewModel (VM) layer: hooks that contain the presentation logic, consume `services/api` and feed the view. It is the functional equivalent of client-side "controllers".
`front-end/Web/src/services/api`	Data access layer: one module per microservice; centralizes REST calls towards Kong.
`front-end/Web/src/navegation`	Definition of public/authenticated navigators (React Navigation) and deep linking; it does not use a traditional web router.
`front-end/Web/src/context` and `core`	Global state (session, theme, responsive) and cross-cutting infrastructure (local storage, constants, utilities).
`front-end/Mobile/src/models`	Data types typed by the 8 bounded contexts, a contractual mirror of the backend DTOs.
`back-end/01..09-ms-*`	One directory per bounded context. Internally: `domain` (entities/ports) → `application` (use cases) → `infrastructure/adapter` (HTTP, persistence, messaging). This is where the business logic lives, equivalent to "backend/controllers" but segregated by hexagonal layers.
`back-end/99-api-gateway/kong`	`kong.yml`: DB-less declaration of upstream services, routes and plugins (rate limiting with Redis, CORS). The only point of public exposure.
`back-end/*.md` and `back-end/quality`	Formal quality documentation: ISO/IEC 25010, ISTQB and ISO 29110 reports, IEEE 829 test specification and catalog of quality codes.
`database/<NN>-ms-*-db/01-ddl…05-rollbacks`	SQL organized by sublanguage (DDL/DML/DCL/TCL) with explicit rollbacks per script: allows each context's schema to be reproduced and reverted.
`database/<NN>-ms-*-db/changelog`	Liquibase changelogs executed automatically when the stack starts (database-as-code).
`database/faceattend_edu_mr_v4.dbml`	Diagram/relational model of the system in DBML as the documentary source of the schema.

Folder	What it does
`tools/seed-api-test-data`	Utility that populates end-to-end test data by consuming the real API (it validates contracts, it does not insert SQL directly).

Source. Own elaboration.

8. Database

8.1 Engines used

The system uses *polyglot persistence*: each technology covers the type of data for which it is best suited.

Table 7

8.1 Engines used

Engine	Version	Use	Container / Name
PostgreSQL	17-alpine	Transactional relational data. One schema per bounded context (*database-per-context* strategy on a single instance).	`faceattend_db`
MongoDB	7	Facial and fingerprint embeddings, in two document collections.	`faceattend_biometric`
Redis	7-alpine	Kong rate-limiting counters (stores no business data).	—
Apache Kafka (KRaft)	3.8.0	Domain event bus (transports events only; persists no business data).	—

Source. Own elaboration.

Connection details (development environment): internal host `postgres` (Docker network `data`), port *5432*. The port is *not published to the client*: only the microservices on the `app` network can reach it. Each service connects with its *own database user*, with privileges only on its own schema (see Section 9.5).

8.2 Database organization (database-per-context)

A single PostgreSQL server hosts *eight schemas*, one per bounded context (PostgreSQL Global Development Group, n.d.). This design makes it possible to migrate each context to its own server in the future without redesigning the model.

Table 8

8.2 Database organization (database-per-context)

Schema	Owning microservice	Tables	Content
`identity`	01-ms-identity	4	People, application users, sessions, and the password policy
`authorization`	02-ms-authorization	4	Roles, permissions, and their assignment to roles and users
`academic`	03-ms-academic	8	Schools, programs, periods, cohorts, courses, academic actors, and enrollments
`scheduling`	04-ms-scheduling	3	Environments, schedule blocks, and class sessions
`attendance`	05-ms-attendance	5	Attendance records, justifications, supporting documents, and reports
`biometric`	06-ms-biometric	0	No relational tables. This context persists exclusively in MongoDB (Section 8.4)
`configuration`	07-ms-configuration	3	Academic and security parameters, and biometric update cases
`notification`	08-ms-notification	2	Alert catalog and raised alerts

Source. Own elaboration.

Design rules of the model:

- *No foreign keys between schemas.* Only logical identifiers travel between contexts; for example, `attendance.attendance_record.academic_actor_id` references `academic.academic_actor` with no physical FK. Integrity between contexts is guaranteed by the application layer. The relational model file marks each of these references explicitly as *cross-context (sin FK)*.
- *Primary keys are chosen per table, not uniform.* `uuid` is used where the identifier travels between contexts or is exposed publicly (`person`, `app_user`, `user_session`, `attendance_report`, `biometric_update_case`); `int`, `smallint`, or `bigint` identity columns are used for catalogs and for high-volume tables; and the two join tables of the `authorization` schema use a *composite primary key*.

- *Standard auditing on every table*, with seven columns: `created_at`, `updated_at`, `deleted_at` (`TIMESTAMP`), `created_by`, `updated_by`, `deleted_by` (`UUID`) and `row_version` (`BIGINT`, default 1) for *optimistic locking*. Deletion is always logical (*soft-delete*).
- *Uniqueness scoped to the owner.* Codes are unique within their school or program rather than globally: `uq_program_code_school`, `uq_course_code_program`, `uq_environment_code_school`, `uq_period_name_school`. The e-mail of a person is unique only among live rows (`where: deleted_at IS NULL`), so a soft-delete releases the address for reuse.
- *Database-as-code:* the schema is created with versioned Liquibase changelogs under `database/<NN>-ms-*-db/01-ddl`, organized into `00-extensions`, `01-schemas`, `02-types`, `03-tables`, `05-materialized-views`, `06-functions`, `07-procedures`, `08-triggers` and `09-indexes`, each with its rollback. Every service, notification included, migrates with Liquibase.

8.3 Entity-relationship diagrams

The complete model is `database/faceattend_edu_mr_v4.dbml`, which can be visualized at dbdiagram.io and is the source of truth for names and types. It is presented below in five figures, one per group of contexts. *Solid* lines are physical foreign keys inside a schema; *dotted* lines are logical references between schemas, which carry no FK.

Figure 2

8.3 Entity-relationship diagrams



Source. Own elaboration.

Figure 3

8.3 Entity-relationship diagrams



Source. Own elaboration.

Figure 4

8.3 Entity-relationship diagrams



Source. Own elaboration.

Figure 5

8.3 Entity-relationship diagrams



Source. Own elaboration.

Figure 6

8.3 Entity-relationship diagrams



Source. Own elaboration.

The `configuration` schema has no diagram of its own: its three tables hold parameters and cases that carry no foreign keys between them. They are documented in Section 8.5.

Key relationships:

Table 9

8.3 Entity-relationship diagrams

Relationship	Cardinality	Meaning
`person` → `app_user`	1:1	A person has at most one application account; `app_user.person_id` is unique.
`app_user` → `user_session`	1:N	Each login opens a session row with its start, end, source IP, and status.
`role` → `role_permission` ← `permission`	N:M	Basis of RBAC access control, with a composite primary key.
`app_user` ⇢ `user_role` → `role`	N:M	A user holds several roles. The reference to the user is logical, since it crosses from `authorization` into `identity`.
`school` → `program` → `course`	1:N	A school offers programs and each program contains courses.
`program` + `academic_period` → `cohort`	N:1 each	A cohort is a group of a program framed within a period.
`person` ⇢ `academic_actor`	1:N (logical)	The same person may act as a student in one school and as an instructor in another; `academic_actor_type` distinguishes the role.
`academic_actor` → `enrollment` ← `cohort`	N:M	An actor enrolls in cohorts; the pair is unique.
`environment` → `schedule_block` → `class_session`	1:N	A recurring block in an environment generates dated sessions; the pair block-date is unique.
`class_session` ⇢ `attendance_record` ⇠ `academic_actor`	1:N (logical)	One record per actor and session, enforced by `uq_attendance_session_actor`.
`attendance_record` → `justification`	1:1	A record admits at most one justification; `attendance_record_id` is unique in `justification`.
`justification` → `supporting_document`	1:N	Evidence files attached to the justification.
`alert_type` → `alert`	1:N	The catalog defines severity and channel; each alert points to its type.

Source. Own elaboration.

Note. The attendance flow therefore crosses three schemas without a single physical foreign key between them: the session lives in `scheduling`, the actor in `academic`, and the record in `attendance`.

8.4 MongoDB collections (biometric embeddings)

The `biometric` context has no relational tables. It persists in two separate document collections, one per biometric modality.

Table 10

8.4 MongoDB collections (biometric embeddings)

Collection	Field	Type	Description
`facial_embedding`	`person_id`	string (UUID)	Owner of the template; the logical reference to `identity.person`.
	`template_version`	int	Version of the template, which allows re-enrollment without losing history.
	`encoding`	array<double>	Feature vector extracted from the face.
	`model_version`	string	Version of the model that produced the vector.
	`enrolled_at`	date	Enrollment date.
	`is_active`	boolean	Marks the template currently in force.
`fingerprint_embedding`	`person_id`	string (UUID)	Owner of the template.
	`finger_number`	int	Finger enrolled, 1 to 10.
	`template_version`	int	Version of the template.
	`encoding`	array<double>	Feature vector extracted from the fingerprint.
	`model_version`	string	Version of the model that produced the vector.
	`enrolled_at`	date	Enrollment date.
	`is_active`	boolean	Marks the template currently in force.

Source. Own elaboration.

The link from the relational side is `configuration.biometric_update_case.current_embedding_ref`, which stores the

document identifier as free text. It is deliberately *not* a real foreign key, because the reference crosses paradigms from SQL to NoSQL.

Embeddings *are not images*: the original face or fingerprint cannot be reconstructed from the vector, which reduces the privacy risk in the event of a leak.

8.5 Data dictionary

Every table listed below additionally carries the seven *audit columns* described in Section 8.2 (`created_at`, `updated_at`, `deleted_at`, `created_by`, `updated_by`, `deleted_by`, `row_version`), which are not repeated in each entry.

Schema `identity`

Table 11

8.5 Data dictionary

Table	Primary key	Business columns	Purpose
`person`	`person_id` uuid	`document_number`, `document_type`, `name`, `last_name`, `email`, `phone`, `blood_type`, `birth_date`, `address`, `status`	Natural person. Unique by document type and number; the e-mail is unique among live rows.
`app_user`	`user_id` uuid	`person_id` (unique), `username` (unique), `password_hash`, `authentication_type`, `status`, `last_access`	Login account. `authentication_type` accepts Local, Windows, or External.
`user_session`	`session_id` uuid	`user_id`, `start_date`, `end_date`, `source_ip`, `session_status`	Session opened by a user. `session_status` is Active or Closed.
`password_policy`	`policy_id` int	`min_length`, `max_length`, `requires_uppercase`, `requires_numbers`, `requires_symbols`, `expiration_days`	Password rules applied at registration and at change.

Source. Own elaboration.

Schema `authorization`

Table 12

8.5 Data dictionary

Table	Primary key	Business columns	Purpose
role	role_id int	role_name (unique), description	Role of the RBAC model.
permission	permission_id int	permission_name (unique), description	Individual permission.
role_permission	(role_id, permission_id)	assignment_date	Permissions granted to a role.
user_role	(user_id, role_id)	assignment_date	Roles assigned to a user. user_id is a logical reference to identity.app_user.

Source. Own elaboration.

Schema `academic`

Table 13

8.5 Data dictionary

Table	Primary key	Business columns	Purpose
school	school_id int	code (unique), name, city_id, city_name, district, country, address, phone, email, status	Training center. city_id points to an external catalog and carries no FK.
program	program_id int	school_id, code, name, status	Training program. The code is unique within its school.
academic_period	academic_period_id int	school_id, name, starts_on, ends_on, is_active	Period over which reports are consolidated.
cohort	cohort_id bigint	program_id, academic_period_id, code (unique), status	Training group of a program within a period.
course	course_id int	program_id, code, name, credit_hours, status	Course or competency. The code is unique within its program.
academic_actor_type	actor_type_id smallint	code (unique), name	Catalog of actor types: STUDENT and INSTRUCTOR.
academic_actor	academic_actor_id bigint	person_id, actor_type_id, school_id, actor_code, started_on, ended_on, status	A person acting in a school under a given type. Unique by school, type, and code.
enrollment	enrollment_id bigint	academic_actor_id, cohort_id, enrolled_on, enrollment_status	Enrollment of an actor in a cohort. Status: Active, Withdrawn, or Completed.

Source. Own elaboration.

Schema `scheduling`

Table 14

8.5 Data dictionary

Table	Primary key	Business columns	Purpose
`environment`	`environment_id` int	`school_id`, `code`, `name`, `capacity`, `status`	Classroom or environment. The code is unique within its school.
`schedule_block`	`schedule_block_id` bigint	`cohort_id`, `course_id`, `environment_id`, `instructor_actor_id`, `day_of_week`, `starts_at`, `ends_at`	Recurring slot. Two unique indexes prevent an environment or an instructor from being double-booked in the same slot.
`class_session`	`class_session_id` bigint	`schedule_block_id`, `session_date`, `opened_by`, `opened_at`, `closed_by`, `closed_at`, `session_status`	Dated instance of a block. Status: Open, Closed, or Cancelled. Unique by block and date.

Source. Own elaboration.

Schema `attendance`

Table 15

8.5 Data dictionary

Table	Primary key	Business columns	Purpose
`attendance_record`	`attendance_record_id` bigint	`class_session_id`, `academic_actor_id`, `attendance_status`, `captured_at`, `capture_method`, `match_score`	Attendance of one actor in one session. Status: Present, Absent, Late, or Justified. Method: FACIAL, MANUAL, IOT, or IMPORT.
`justification_type`	`justification_type_id` int	`school_id`, `name`, `description`, `requires_attachment`, `status`	Catalog of justification types. A null `school_id` marks a global type.
`justification`	`justification_id` bigint	`attendance_record_id` (unique), `justification_type_id`, `reason`, `submitted_at`, `reviewed_by`, `reviewed_at`, `review_status`, `resolution_notes`	Justification of an absence. Status: Pending, Approved, or Rejected.
`supporting_document`	`supporting_document_id` bigint	`justification_id`, `file_name`, `storage_uri`, `mime_type`, `size_bytes`	Evidence file attached to a justification.

Table	Primary key	Business columns	Purpose
attendance_report	report_id uuid	school_id, cohort_id, generated_by, report_type, filters (jsonb), result (jsonb), generated_at	Generated report, with its filters and its result stored as JSON.

Source. Own elaboration.

Schema `configuration`

Table 16

8.5 Data dictionary

Table	Primary key	Business columns	Purpose
academic_configuration	configuration_id int	school_id, configuration_name, configuration_value, description	Academic parameter per school, for example tardy_tolerance_minutes. Unique by school and name.
security_configuration	configuration_id int	configuration_name (unique), configuration_value, description	Global security parameter.
biometric_update_case	case_id uuid	person_id, biometric_type, finger_number, current_embedding_ref, reason, update_status, requested_by, requested_at, reviewed_by, reviewed_at, resolution_notes	Request to re-enroll a biometric template. A CHECK constraint requires finger_number between 1 and 10 when the type is FINGERPRINT and null when it is FACIAL.

Source. Own elaboration.

Schema `notification`

Table 17

8.5 Data dictionary

Table	Primary key	Business columns	Purpose
alert_type	alert_type_id smallint	code (unique), name, severity, channel	Catalog of alerts. Severity: INFO, WARNING, or CRITICAL. Channel: DASHBOARD, EMAIL, or PUSH.
alert	alert_id bigint	academic_actor_id, alert_type_id, raised_at, resolved_at	Alert raised for an actor and its resolution.

Source. Own elaboration.

8.6 Divergences between the model and the implemented schema

The relational model file and the Liquibase changelogs do not cover exactly the same set of tables (Liquibase, n.d.). The difference matters when reading the schema, so it is recorded here.

Table 18

8.6 Divergences between the model and the implemented schema

Divergence	Detail
`identity.city`	The changelogs still create this table; the model file removed it in revision v6, when the city catalog moved to an external API consumed through `school.city_id` and `school.city_name`.
Quality tables	The `configuration` changelogs additionally create `quality_project`, `quality_evaluation`, `quality_evaluation_item`, `process_assessment`, `process_assessment_rating`, `istqb_assessment` and `istqb_assessment_item`, which support the reports of `09-ms-quality` and are not part of the relational model file.
Totals	The model file declares 29 tables; the changelogs create 37, the eight above being the difference.

Source. Own elaboration.

Before relying on a name or a type, check `database/faceattend_edu_mr_v4.dbml` for the modelled tables and the changelogs under `database/<NN>-ms-*-db/01-ddl/03-tables` for what is actually created.

9. Security

Security is applied in *layers* (defense in depth): network, gateway, authentication, authorization, data and auditing.

9.1 Authentication (opaque session)

1. The client sends `{identifier, password}` to `POST /api/v1/auth/login` through Kong (public route).
2. *ms-identity* looks up the user and compares the password against the stored *BCrypt* hash. On failure it answers 401 with the same message for an unknown user and a

wrong password.

3. If it is valid, it inserts a row in `identity.user_session` and answers `{sessionId, userId, sessionStatus, startDate, endDate, sourceIp}`. The `sessionId` is an *opaque UUID*: it is not a JWT and carries no *claims* or signature.
4. The client sends the `sessionId` on every request: `Authorization: Bearer <sessionId>`.
5. *Kong* (global Lua pre-function) rejects with 401 any non-public route without `Bearer <UUID>` and validates the session with `GET /api/v1/sessions/{id}` against ms-identity (10 s cache). It does not inject identity headers.
6. *Each microservice* validates again: session against ms-identity (`/api/v1/sessions/{id}`) and permissions against ms-authorization (`/api/v1/users/{id}/roles` and `GET /api/v1/auth/evaluate?userId&permission`).

Table 19

9.1 Authentication (opaque session)

Parameter	Value
Token type	Opaque UUID `sessionId` (row of `identity.user_session`)
JWT library	None in the backend
Session lifetime	480 minutes (`session_timeout_minutes`); lazy expiry when the session is read
Renewal	`POST /api/v1/auth/refresh {sessionId}`: rotates the session (new `sessionId`, the old one is closed); single use
Logout	`POST /api/v1/auth/logout?sessionId=<UUID>` (204); Kong drops its cache entry for that session
Per-service guards	Java: `AuthTokenFilter` · TypeScript: `authGuard.ts` · Python: `security.py`
Client-side storage	Web: browser local storage · Mobile: `expo-secure-store` (keystore/Keychain); AsyncStorage only as the web fallback

Source. Own elaboration.

Why an opaque session: revocation is immediate (close the `user_session` row), there are no keys to guard or rotate and no service can forge sessions. The cost is one ms-identity lookup per request, softened by short caches (10 s) in Kong and in the services.

Future evolution (ADR-007)

ADR-007 considers, as an improvement not yet implemented, having ms-identity issue a *JWT signed with RS256* (private key only in ms-identity, verification with the public key in the other services, for example with the `jjwt` library) to avoid the session lookup on every request. Until it is adopted, nothing in the system issues or validates JWTs.

9.2 Password encryption

- Passwords are stored with *BCrypt* (an adaptive hash function with a random per-user *salt*), with cost factor *12* and version `$2b` (ms-identity `SecurityConfig`).
- The plain-text password is never stored or written to logs; the `password_hash` field is never returned in API responses.
- The hash is not reversible: to validate, the supplied value is hashed and compared.

9.3 Role-based authorization (RBAC)

- *ms-authorization* defines the roles (*ADMIN, TEACHER, STUDENT*) and the permissions associated with each one (`roles` ↔ `permissions` model, section 8.3).
- Each microservice validates the session against ms-identity and asks ms-authorization whether the user holds the permission the operation requires: `AuthTokenFilter` in Java, `authGuard.ts` in TypeScript and `security.py` in Python.

Table 20

9.3 Role-based authorization (RBAC)

Role	Scope (summary)
ADMIN	Management of users, roles, programs, courses, configuration and querying of quality metrics.
TEACHER	Open/close sessions of their own courses and query attendance for their own courses.
STUDENT	Record their own attendance and query only their own history.

Source. Own elaboration.

9.4 Measures preventing a user from accessing other people's data

Table 21

9.4 Measures preventing a user from accessing other people's data

Measure	How it is applied
Role-based authorization	Every endpoint requires a minimum role/permission; a `STUDENT` cannot invoke `TEACHER` or `ADMIN` operations.
Owner-based filtering	Queries take the identity from the `userId` of the validated session (not from a parameter the client can send), so a student only sees their own records.
A single entry point	The client can only reach Kong; the microservices and the databases are not accessible from outside.
Network segmentation into 3 zones	`edge` (client ↔ Kong), `app` (Kong ↔ microservices) and `data` (microservices ↔ databases only). A client cannot reach the `data` network.
Rate limiting (Kong + Redis)	Limits the number of requests per consumer/IP, mitigating brute force and API abuse.
Restricted CORS	Kong accepts only authorized origins (the web frontend).
Input validation	DTO validation in every service (Bean Validation in Java, schemas in Fastify (zod), Pydantic in FastAPI).
Parameterized queries	JPA and the `pg` driver use *prepared statements*, preventing SQL injection.
Data isolation per service	Each microservice only has credentials for its own schema; it cannot read tables of another context.
Soft-delete and auditing	Records are not physically deleted (`deleted_at`) and every change retains its author and date (`created_by`, `updated_by`).

Source. Own elaboration.

9.5 Database security

- *Per-service users (DCL):* the scripts in `database/<NN>-ms-*-db/03-dcl` create one database role per microservice with least privileges (`SELECT/INSERT/UPDATE/DELETE` only on its own schema; no `SUPERUSER` and no `CREATE DB`).
- *Separate migration user:* Liquibase uses its own user with DDL privileges; the running services do *not* have DDL privileges.
- *Secrets outside the code:* DB passwords, internal secrets (e.g. `EMAIL_INTERNAL_TOKEN`) and credentials are injected by means of *environment*

variables / an `.env` *file* (excluded from version control with `.gitignore`).

- *Biometric data:* only *vectors (embeddings)* are stored, not images of the face or the fingerprint.
- *Concurrent integrity:* the `version` field (*optimistic locking*) prevents simultaneous overwrites.

10. Maintenance

10.1 Backup and restoration

Table 22

10.1 Backup and restoration

Element	What to back up	Tool	Recommended frequency
PostgreSQL	All schemas of `faceattend_edu`	`pg_dump` (custom format)	Daily (full)
MongoDB	`biometric_embeddings` collection	`mongodump`	Daily
`.env` file	Environment variables and secrets	Encrypted copy in secure storage	On generation / rotation
Code and scripts	Repository (includes `database/` with DDL and rollbacks)	Git + GitHub	On every *push*

Source. Own elaboration.

Steps to back up PostgreSQL (Docker Compose):

```
# 1. Generate the backup from the container
docker compose exec -T postgres pg_dump -U <admin_user> -Fc faceattend_edu >
backup_$(date +%F).dump

# 2. Verify that the file is not empty and copy it to an external location
ls -lh backup_$(date +%F).dump
```

Steps to back up MongoDB:

```
docker compose exec -T mongo mongodump --db faceattend_biometric --archive --gzip >
mongo_$(date +%F).gz
```

Steps to restore:

```
# PostgreSQL
docker compose exec -T postgres pg_restore -U <admin_user> -d faceattend_edu --clean <
backup_YYYY-MM-DD.dump
```

```
# MongoDB
docker compose exec -T mongo mongorestore --archive --gzip < mongo_YYYY-MM-DD.gz
```

Good practices:

- Apply the *3-2-1 rule*: 3 copies, on 2 different media, 1 off-site.
- *Test the restoration* periodically in a separate environment; an unverified backup is not a backup.
- Keep a *retention* of at least *30 days* of daily backups.
- Back up PostgreSQL and MongoDB at the *same moment*, since `biometric_templates.mongo_ref` links the two.
- Schema changes are reverted with the scripts in `05-rollbacks` or with `liquibase rollback`, without needing to restore a full backup.

10.2 Dependency updates and vulnerabilities

Table 23

10.2 Dependency updates and vulnerabilities

Stack	Services	Review command	Action
Node.js / TypeScript	ms-academic, ms-configuration, ms-quality, frontends	`npm audit` · `npm outdated`	`npm audit fix` / `npm update`
Java / Maven	ms-identity, ms-authorization, ms-scheduling, ms-attendance	`./mvnw versions:display-dependency-updates`	Update versions in `pom.xml`
Python / Poetry	ms-biometric	`poetry show --outdated`	`poetry update`
Docker images	The whole stack	`docker compose pull`	Rebuild with `docker compose build --no-cache`

Source. Own elaboration.

Recommendations:

- Review vulnerabilities *at least once a month* and always before a delivery.

- Update *patch/minor* versions first; *major* ones (e.g. Spring Boot, Expo) require testing the complete system with the ms-quality test suite (VMware, n.d.).
- After every update: run the tests, bring the stack up with `docker compose up --build` and validate the *healthchecks*.
- *Rotate the `.env` secrets* (DB passwords, `EMAIL_INTERNAL_TOKEN`, SMTP) periodically (*every 6 months* or on suspicion of compromise). To end every active session, close them in `identity.user_session`.
- Change the default passwords of the development environments before any real deployment.

10.3 Monitoring and support

- *System status:* `docker compose ps` and the *healthchecks* of each container; *ms-quality* exposes verification metrics.
- *Logs:* `docker compose logs -f <service>` (the Node services log with pino, the Java ones with Spring Boot logging and the Python one with structlog).
- *Error table:* the `back-end/quality/quality-codes.md` catalog makes it possible to quickly identify the origin of a failure.
- *Incidents:* record incidents using the IEEE 829 format (`back-end/*incidencias*.md`).
- *Technical support:* report problems by means of *issues* in the GitHub repository, attaching logs, the version (*1.0.0*) and steps to reproduce.

11. Programming interface (API)

11.1 General conventions

The whole API is exposed through *Kong* at a single entry point (Kong Inc., n.d.). Clients never invoke a microservice directly.

Table 24

11.1 General conventions

Aspect	Convention
Entry point	`http://<host>:8000` (Kong proxy port)
Version prefix	`/api/v1/...` — every valid route begins with this prefix
Exchange format	JSON (`Content-Type: application/json`)
Authentication	`Authorization: Bearer <sessionId>` header; the public `/api/v1/auth` routes (login, login/verify, refresh, logout, forgot-password, verify-code, reset-password) and `POST /api/v1/onboarding/schools` are the exceptions
Identifiers	UUID v4 on all resources; never auto-incrementing integers
Dates and times	ISO 8601 with time zone (`timestamptz`), for example `2026-10-07T08:30:00-05:00`
Pagination	`page` (zero-based) and `size` parameters; the response includes `totalElements` and `totalPages`
Filtering and sorting	Query parameters per field and `sort=<field>,<asc|desc>`
Deletion	Logical (`DELETE` sets `deleted_at`); there is no physical deletion from the API
Concurrency	`version` field in the body of update operations (*optimistic locking*)

Source. Own elaboration.

11.2 Verbs and status codes

Table 25

11.2 Verbs and status codes

Verb	Use	Success code
`GET`	Query one or several resources	`200 OK`
`POST`	Create a resource	`201 Created` (`Location` header)
`PUT`	Full replacement of the resource	`200 OK`
`PATCH`	Partial update	`200 OK`
`DELETE`	Logical deletion	`204 No Content`

Source. Own elaboration.

Table 26

11.2 Verbs and status codes

Code	Meaning in the system
`400 Bad Request`	DTO validation failed (missing field, invalid format).
`401 Unauthorized`	`sessionId` missing, malformed, closed or expired. Issued by Kong at the edge and, if the request gets through, also by the destination microservice's guard.
`403 Forbidden`	Valid session but the role does not hold the required RBAC permission.
`404 Not Found`	The resource does not exist or has been logically deleted.
`409 Conflict`	Uniqueness violation (duplicate email or document number) or version conflict.
`422 Unprocessable Entity`	Business rule breached (e.g. recording attendance in a closed session).
`429 Too Many Requests`	Kong rate limit exceeded (counter in Redis).
`500 Internal Server Error`	Unhandled error; logged with a correlation identifier.
`502 Bad Gateway`	The destination microservice is not yet *healthy*.
`503 Service Unavailable`	Dependency down (database or Kafka).

Source. Own elaboration.

11.3 Error format

All services return errors with the same structure, so that the client can handle them uniformly:

```
{
  "timestamp": "2026-10-07T08:30:00-05:00",
  "status": 409,
  "code": "IDENTITY-409-002",
  "message": "The document number is already registered",
  "path": "/api/v1/persons",
  "correlationId": "7f3a9c54-0c21-4f0e-9b2a-1d8e6f4b5c77"
}
```

The `code` field follows the pattern `<CONTEXT>-<HTTP>-<SEQUENCE>` and is cataloged in `back-end/quality/quality-codes.md`, which is the source of truth for interpreting failures.

11.4 Authentication and session renewal

Table 27

11.4 Authentication and session renewal

Endpoint	Verb	Description
`/api/v1/auth/login`	`POST`	Receives `identifier` and `password`; returns `{sessionId, userId, sessionStatus, startDate, endDate, sourceIp}`. Public.
`/api/v1/auth/refresh`	`POST`	Receives a valid `{sessionId}`; returns a new session and closes the old one (single use). Public.
`/api/v1/auth/logout`	`POST`	`?sessionId=<UUID>`; closes the session (204). Public.
`/api/v1/auth/evaluate`	`GET`	Exposed by ms-authorization; `?userId&permission` answers whether the user holds that permission.

Source. Own elaboration.

```
# Login
curl -X POST http://localhost:8000/api/v1/auth/login \
  -H "Content-Type: application/json" \
  -d '{"identifier":"<administrator-user>","password":"<password>"}'

# Using the token on a protected endpoint
curl http://localhost:8000/api/v1/persons \
  -H "Authorization: Bearer <sessionId>"
```

Lifetime: 480-minute session; `refresh` rotates it before it expires (section 9.1).

11.5 Route map by context

Table 28

11.5 Route map by context

Microservice	Main routes
01 identity	`/api/v1/auth`, `/api/v1/persons`, `/api/v1/users`, `/api/v1/sessions`, `/api/v1/cities`, `/api/v1/password-policies`
02 authorization	`/api/v1/roles`, `/api/v1/permissions`, `/api/v1/auth/evaluate`, `/api/v1/users/{id}/roles`
03 academic	`/api/v1/schools`, `/api/v1/programs`, `/api/v1/academic-periods`, `/api/v1/cohorts`, `/api/v1/courses`, `/api/v1/academic-actors`, `/api/v1/enrollments`
04 scheduling	`/api/v1/environments`, `/api/v1/schedule-blocks`, `/api/v1/class-sessions`
05 attendance	`/api/v1/attendance-records`, `/api/v1/justifications`, `/api/v1/justification-types`, `/api/v1/supporting-documents`
06 biometric	`/api/v1/biometric` (enrollment and verification)
07 configuration	`/api/v1/academic-configurations`, `/api/v1/security-configurations`, `/api/v1/biometric-update-cases`

Microservice	Main routes
08 notification	`/api/v1/alert-types`, `/api/v1/alerts`
09 quality	`/api/v1/quality`, `/health/quality`

Source. Own elaboration.

11.6 Health endpoints

Table 29

11.6 Health endpoints

Technology	Health route
Spring Boot (identity)	`/api/v1/health`
Spring Boot (scheduling, attendance, authorization)	`/actuator/health`
Fastify (academic, configuration, quality)	`/health`
FastAPI (biometric)	`/health` · interactive documentation at `/docs`
Fastify (notification)	`/health`

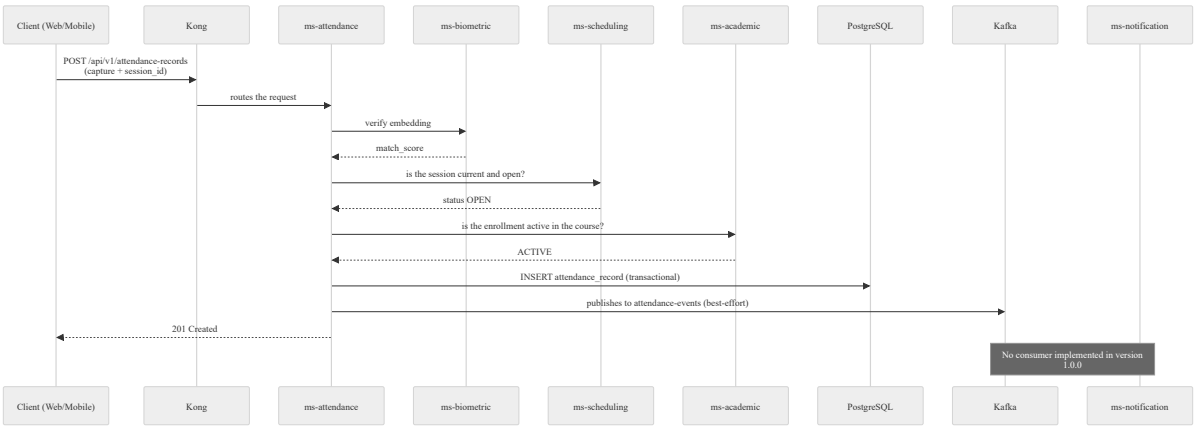
Source. Own elaboration.

The Java services additionally publish their OpenAPI documentation at `/swagger-ui.html`.

11.7 Attendance recording flow (complete sequence)

Figure 7

11.7 Attendance recording flow (complete sequence)



Source. Own elaboration.

11.8 Domain events (Kafka)

Table 30

11.8 Domain events (Kafka)

Topic	Producer	Events it carries
`identity-events`	01 ms-identity	Creation and modification of persons, users, credentials, and sessions
`city-events`	01 ms-identity	Changes to the city catalog
`password-policy-events`	01 ms-identity	Changes to the password policies
`role-events`	02 ms-authorization	Creation, modification, and assignment of roles
`permission-events`	02 ms-authorization	Changes to permissions and to the role-permission matrix
`environment-events`	04 ms-scheduling	Registration and modification of environments
`schedule-block-events`	04 ms-scheduling	Changes to schedule blocks
`class-session-events`	04 ms-scheduling	Opening, closing, and cancellation of class sessions
`attendance-events`	05 ms-attendance	Attendance records created or adjusted
`justification-events`	05 ms-attendance	Submission and resolution of absence justifications

Topic	Producer	Events it carries
`justification-type-events`	05 ms-attendance	Changes to the justification type catalog
`document-events`	05 ms-attendance	Supporting documents attached to a justification

Source. Own elaboration.

Publication is *best-effort and non-blocking*: the publisher catches any failure, records it as a warning in the log, and lets the transaction complete. An unreachable broker therefore never prevents an attendance record from being written.

Current state of consumption. `ms-notification` is the only consumer: it subscribes to `attendance.justification.events` and `scheduling.class-session.closed` to build alerts, which it also exposes over HTTP and sends by SMTP. The other topics have no consumers yet; they exist so that consumers can be added without modifying the producers.

12. Testing and quality assurance

12.1 Normative framework applied

Table 31

12.1 Normative framework applied

Standard	What it contributes to the project	Evidence in the repository
ISO/IEC 25010	Product quality model: characteristics and subcharacteristics evaluated.	`back-end/ISO25010*.md`, ms-quality metrics
IEEE 829	Test documentation formats: plan, case specification, incident report and summary.	`back-end/IEEE829*.md`
ISTQB	Terminology, test levels and types; case design techniques.	`back-end/ISTQB*.md`
ISO/IEC 29110	Lifecycle profile for Very Small Entities (VSE): project management and software implementation.	`back-end/ISO29110*.md`

Source. Own elaboration.

12.2 Quality characteristics evaluated (ISO/IEC 25010)

Table 32

12.2 Quality characteristics evaluated (ISO/IEC 25010)

Characteristic	How it is addressed in FaceAttend EDU
Functional suitability	Coverage of functional requirements FR1–FR8; validation through IEEE 829 test cases.
Performance efficiency	Response time of the critical endpoints; rate limiting in Kong; indexes in PostgreSQL.
Compatibility	Versioned REST API; Web and Mobile clients from a single code base (Expo).
Usability	MVVM pattern, accessible themes, color-blindness modes and WCAG contrast evaluator in the client.
Reliability	*Healthchecks*, startup retries, logical deletion and *optimistic locking*.
Security	Opaque session validated at Kong and in every service, RBAC, BCrypt, network segmentation into 3 zones, per-service DB users (section 9).
Maintainability	DDD + hexagonal architecture, ADRs, database-as-code, catalog of error codes.
Portability	The whole stack is brought up with Docker Compose; no dependencies on the host system.

Source. Own elaboration.

12.3 Test levels and types (ISTQB)

Table 33

12.3 Test levels and types (ISTQB)

Level	Scope	Tools per stack
Unit	Entities, *value objects* and use cases, without infrastructure.	JUnit 5 + Mockito (Java) · Jest (TS) · pytest (Python, `tests/unit`)
Integration	Persistence and messaging adapters against real engines.	Spring Boot Test · Testcontainers ^10.4.0 (declared in `03-ms-academic`, not yet used by any test) · pytest with fixtures (`tests/support`)
Component / API	Contract of each microservice through its HTTP port.	Postman/Insomnia · `curl` · verification collections
System (E2E)	Complete flow through Kong, with real API data.	`tools/seed-api-test-data` (loads via API, validates contracts)
Acceptance	Verification of the FRs by the assessing instructor.	IEEE 829 cases with documented expected result

Source. Own elaboration.

Test types applied: functional (positive and negative), security (authentication, authorization, network isolation — test 5 of the installation manual), basic performance,

regression after every deployment, and database migration (application and *rollback*).

12.4 Test documentation (IEEE 829)

Table 34

12.4 Test documentation (IEEE 829)

Document	Content	File
Test plan	Scope, strategy, entry and exit criteria, risks, schedule.	`back-end/IEEE829-plan*.md`
Case specification	Identifier, preconditions, data, steps, expected result, traceability to the FR.	`back-end/IEEE829-casos*.md`
Incident report	Identifier, severity, priority, steps to reproduce, evidence, status.	`back-end/IEEE829-incidencias*.md`
Test summary	Executed, passed, failed, blocked; conclusion on the release.	`back-end/IEEE829-resumen*.md`

Source. Own elaboration.

Exit criterion: a version is not released with *critical* or *major* severity incidents open, nor with the migration chain in a failed state.

12.5 The quality microservice (`09-ms-quality`)

`ms-quality` does not take part in the business: it is a *verification instrument* that exposes objective metrics about the system.

Table 35

12.5 The quality microservice (09-ms-quality)

Function	Detail
Metrics consolidation	Exposes ISO/IEC 25010 indicators computed over the real state of the stack.
Availability verification	`/health/quality` through Kong confirms the complete gateway → service chain.
Quality reporting	Delivers the data that feeds the ISO 25010, ISTQB, IEEE 829 and ISO 29110 reports.
Code catalog	Cross-reference with `back-end/quality/quality-codes.md` for interpreting failures.

Source. Own elaboration.

12.6 Test data

Data loading is *always* performed through the real API, never with a direct `INSERT`:

```
cd tools/seed-api-test-data
npm run seed
```

This approach validates the API contracts in the very act of populating the system and is *idempotent*: it detects pre-existing records and does not duplicate them.

13. Deployment and environments

13.1 Planned environments

Table 36

13.1 Planned environments

Environment	Purpose	Characteristics
Local / development	Day-to-day work of the team and assessment.	Complete Docker Compose; development credentials; `BIND_IP=127.0.0.1`.
Testing / demonstration	Functional validation and showcasing of the system.	Same stack with seed data loaded via the API; only the necessary ports open.
Production	Outside the current scope (section 3).	Would require a public domain, TLS, high availability and a secrets manager.

Source. Own elaboration.

13.2 Docker network topology

Table 37

13.2 Docker network topology

Network	Who belongs to it	Who can reach it
`faceattend-edge`	`frontend-web`, `kong-gateway`	The user's browser.
`faceattend-app`	`kong-gateway`, `redis`, the 9 microservices	Only Kong and the microservices among themselves.
`faceattend-data`	`postgres`, `mongodb`, `kafka`, the 8 migrations	Only the `app` layer. Unreachable from the client.

Source. Own elaboration.

This separation is a *verifiable security control*: the fifth test of the installation manual checks that `frontend-web` cannot resolve the name `postgres`.

13.3 Startup chain and healthchecks

Table 38

13.3 Startup chain and healthchecks

Phase	Components	Condition for advancing
1	`postgres`	Responds to its *healthcheck* (`pg_isready`).
2	8 Liquibase containers (identity → authorization → academic → scheduling → attendance → biometric → configuration → notification)	Each migration finishes with exit code 0.
3	The 9 microservices	Each one waits for its own migration and for a healthy DB.
4	`kong-gateway` + `redis`	Waits for Redis, Kafka and the 8 main microservices.
5	`frontend-web`	Waits for Kong to be healthy.

Source. Own elaboration.

If a migration fails, the chain stops: *the migration logs are always the first place to look*.

13.4 Relevant environment variables

The complete configuration resides in a single `.env` file at the root of `FULL/`. The installation manual documents all of the variables; those with the greatest technical impact are:

Table 39

13.4 Relevant environment variables

Variable	Technical effect
`BIND_IP`	Binding address of the internal ports. `127.0.0.1` keeps the data layer inaccessible from the local network.
`EXPO_PUBLIC_API_URL`	Is baked into the bundle during the build; changing it requires `docker compose up -d --build frontend-web`.
`POSTGRES_*` / `MONGO_*`	Credentials and ports of the persistence engines.
`KONG_PROXY_PORT` / `KONG_ADMIN_PORT`	Public API port and administration port (local).

Variable	Technical effect
FACEATTEND_LIQUIBASE_IMAGE	Image that runs the migrations; pins the Liquibase version.
BOOTSTRAP_* / SEED_*	Initial users created by the migrations and the user of the seeding tool.

Source. Own elaboration.

Rule: the `.env` file is excluded from version control (`.gitignore`) and must never be published. The passwords and internal secrets are injected by environment variable (section 9.5).

13.5 Operation commands

```
docker compose up -d --build        # full deployment (first time: 15–40 min)
docker compose ps -a                # status and health of the 25 containers
docker compose logs -f <service>     # live logs
docker compose restart <service>     # targeted restart
docker compose config                # configuration validation
docker compose down                  # stops, preserving volumes
docker compose down -v               # DESTRUCTIVE: also removes the data
```

13.6 Observability

Table 40

13.6 Observability

Element	Implementation
Structured logs	pino in the Node (Fastify) services, Spring Boot logging in Java and structlog in FastAPI.
Traces	No service is instrumented with distributed tracing (OpenTelemetry); it is pending work.
Health metrics	`/actuator/health`, `/health` and `/health/quality`; `docker compose ps` for the aggregate status.
Correlation	Correlation identifier propagated in the header and recorded in the error body (section 11.3).

Source. Own elaboration.

14. Development conventions and version control

14.1 Repository structure

Monorepo with three large areas (`front-end/`, `back-end/`, `database/`) plus `tools/`. Each microservice is autonomous: its own `Dockerfile`, its own dependency manager and its own database schema.

14.2 Mandatory internal architecture

Every new service respects the *hexagonal architecture* described in section 6.2:

- `domain/` — entities, *value objects* and *ports* (interfaces). No external dependencies.
- `application/` — use cases that orchestrate the domain.
- `infrastructure/` or `adapter/` — HTTP, persistence and messaging adapters.

Dependency rule: the outer layers depend on the inner ones, never the other way round. The domain does not import Spring, Fastify or FastAPI.

14.3 Naming conventions

Table 41

14.3 Naming conventions

Element	Convention	Example
Tables and columns	`snake_case`, plural for tables	`attendance_records`, `match_score`
Schemas	Name of the *bounded context* in the singular	`identity`, `scheduling`
Java / Python classes	`PascalCase`	`AttendanceRecord`
TS/Python variables and functions	`camelCase` / `snake_case` depending on the language	`findActiveEnrollment`
API routes	`kebab-case`, plural	`/api/v1/attendance-records`
Error codes	`<CONTEXT>-<HTTP>-<SEQUENCE>`	`IDENTITY-409-002`
SQL scripts	`<NN>-<type>-<description>.sql` inside `01-ddl` … `05-rollbacks`	`01-ddl/01-create-users.sql`

Source. Own elaboration.

The complete database conventions are in `database/CONVENCIONES.md`.

14.4 Version control

Table 42

14.4 Version control

Aspect	Project practice
Tool	Git 2.39.5, remote on GitHub

Aspect	Project practice
Working branch	`develop` (the reference branch of this manual)
Branches per cycle	One branch per integration cycle; `main` reserved for delivered versions
Commit messages	Conventional commits: `feat(ms-attendance): …`, `fix(ms-identity): …`, `chore(compose): …`, `docs(manual): …`
Product versioning	Semantic: `1.0.0` is the documented version
Design decisions	ADRs (Architecture Decision Records), one per decision, with context, decision and consequences

Source. Own elaboration.

14.5 Database as code

No schema change is applied by hand:

1. The script is written in the corresponding folder (`01-ddl`, `02-dml`, `03-dcl`, `04-tcl`).
2. Its *rollback* is written *mandatorily* in `05-rollbacks`.
3. The *changeset* is registered in the context's `changelog.xml`.
4. Liquibase applies it on the next startup, ms-notification included.
5. The documentary model `database/faceattend_edu_mr_v4.dbml` is updated to reflect the change.

14.6 Criteria for integrating a change

- The service compiles and its unit tests pass.
- `docker compose up -d --build` brings up the complete stack with all *healthchecks* green.
- The 8 migrations finish with exit code 0.
- The five verification tests of the installation manual are passed.
- No foreign keys between schemas are introduced, and no service queries tables of another context.
- No secrets are added to the repository.

- The affected documentation (this manual, `SERVICES.md`, `DATABASE.md`, ADRs) is updated in the same change.

15. Glossary

Table 43

15. Glossary

Term	Definition in the context of the project
ADR	*Architecture Decision Record.* A brief document that records an architectural decision, its context and its consequences.
Hexagonal architecture	Pattern that isolates the domain from the infrastructure by means of ports (interfaces) and adapters.
Bounded context	Boundary of a domain model in DDD. In FaceAttend EDU, each microservice is a bounded context.
BCrypt	Adaptive hash function with a random *salt*, used to store passwords.
Database-per-context	Strategy in which each context owns its own store; here, one PostgreSQL schema per service.
DBML	*Database Markup Language.* Textual language for describing and visualizing the relational model.
DDD	*Domain-Driven Design.* Design approach centered on the domain model.
Embedding	Numeric vector that represents facial or fingerprint features. It does not allow the original image to be reconstructed.
Healthcheck	Periodic check that Docker runs against a container to determine whether it is operational.
sessionId	Opaque UUID identifier of an `identity.user_session` row; sent as `Bearer` and validated against ms-identity.
KRaft	Kafka operating mode that removes the dependency on Zookeeper.
Liquibase	Tool for versioned database migrations (*database-as-code*).
MVVM	*Model–View–ViewModel.* Pattern of the Web and Mobile clients: the view consumes *view models* that encapsulate the presentation logic.
Optimistic locking	Concurrency control by means of the `version` field: a write with a stale version is rejected.
Polyglot persistence	Use of several storage engines, each one for the data type it supports best.
Port / adapter	Interface defined in the domain (port) and concrete implementation in infrastructure (adapter).
RBAC	*Role-Based Access Control.* Permissions are assigned to roles and roles to users.
Rate limiting	Per-consumer request limit applied by Kong with counters in Redis.
Logical reference	Relationship between schemas materialized only by UUID, without a physical foreign key.
Soft-delete	Logical deletion by means of the `deleted_at` marker; the record is preserved for auditing.

Term	Definition in the context of the project
Topic	Kafka channel through which a type of domain event travels.

Source. Own elaboration.

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